FIELD SURVEY REPORT

BENTHIC COMMUNITY SAMPLING OF GULF STURGEON FORAGING AREAS, FALL 2024

Gulf Sturgeon Foraging Sites

The Fall 2024 Gulf sturgeon foraging site benthic sampling was conducted by Thompson Engineering personnel at 79 sampling locations in the waters surrounding Ship Island. Field operations were coordinated by Jonathan O'Neal, Senior Biologist at Thompson Engineering. Figure 1 shows an overall map of the sampling locations for the October-November 2024 sampling event. Figures 2-7 show detailed maps including station names.

Sampling was performed over a time period beginning on October 25, 2024, and ending on November 25, 2024.

Field Collection Methods

Three replicate benthic macroinfaunal samples were collected at each of the 79 sturgeon foraging site sampling stations using a modified Van Veen grab (sampling area of 0.04 m^2), resulting in a total of 237 macroinfaunal samples. Field sampling personnel inspected each replicate to ensure consistent sampling. Any replicates that were inconsistent or in any way suspect (poor sediment volume can be the result of wave action or large shells causing premature or incomplete closing of the grab) were discarded and resampled. One additional benthic sample was collected at each station to be analyzed for sediment texture (grain size) and TOC (79 total sediment samples).

Macroinfaunal samples were rinsed in the field through a 0.5-mm mesh screen and transferred to a labeled 7" x 12.5" geological sample bag (manufactured by Hubco, Inc.). Collected samples were stored in 5-gal. buckets containing a 10% formalin solution for transport to Thompson Engineering's taxonomy laboratory in Mobile. Sediment samples were homogenized and placed in labeled Whir-Pak® bags and stored on ice for transport to the analytical laboratories.

At each station, standard water quality measurements were taken at three depths (near-surface, middle, and near-bottom) prior to benthic sampling. A calibrated YSI® ProDSS datasonde was used to measure temperature, conductivity, salinity, pH and dissolved oxygen (DO) concentration. Table 1 gives the water quality data for all of the sampling sites during the October-November 2024 sampling event.

Deviations From Original Scope of Work

No deviations from the original project plan occurred during the benthic sampling event. All hydrographic measurements and sample collections (number of stations, station locations, and collection procedures) were performed in accordance with the technical scope of work for this project.

Table 1. Water quality data for the benthic community stations during the Fall 2024 sampling event.

Station		DateTime M/D/Y	Depth ft	Temp °C	SpCond mS/cm	Salinity ppt	pH	DO %	DO mg/L
Horne Island Sites									
BI-HN1	B	11/25/24 8:30	17.12	19.19	49.86	32.71	8.02	95.0	7.23
	M	11/25/24 8:31	8.87	18.34	48.40	31.64	8.08	107.2	8.34
	S	11/25/24 8:32	0.73	17.59	46.02	29.91	8.09	109.8	8.76
BI-HN2	B	11/25/24 8:10	13.13	18.85	49.11	32.16	7.98	98.2	7.55
	M	11/25/24 8:12	6.81	18.32	48.14	31.44	8.02	105.2	8.20
	S	11/25/24 8:13	0.72	17.58	47.00	30.61	8.07	109.7	8.72
BI-HS1	B	11/23/24 12:49	20.18	20.21	51.70	34.05	8.19	100.8	7.47
	M	11/23/24 12:50	10.61	20.29	51.69	34.05	8.17	98.8	7.31
	S	11/23/24 12:51	0.69	20.76	51.73	34.08	8.16	99.2	7.27
BI-HS2	B	11/23/24 12:32	16.89	19.22	51.32	33.78	8.19	103.5	7.82
	M	11/23/24 12:33	8.36	19.53	51.35	33.80	8.17	100.7	7.57
	S	11/23/24 12:34	0.69	20.31	51.50	33.91	8.16	99.4	7.36
Dog Key Pass Sites									
BI-DK15	B	11/23/24 13:05	11.31	19.01	51.26	33.73	8.21	109.2	8.29
	M	11/23/24 13:06	5.65	19.47	51.26	33.74	8.18	102.3	7.70
	S	11/23/24 13:07	0.68	19.89	51.20	33.69	8.16	101.5	7.58
BI-NDK4	B	11/24/24 7:59	6.67	18.63	49.57	32.49	8.04	108.3	8.34
	M	11/24/24 8:00	3.47	17.98	48.63	31.81	8.10	107.5	8.42
	S	11/24/24 8:01	0.72	17.91	48.55	31.75	8.13	107.3	8.42
BI-DK13	B	11/24/24 8:16	13.15	18.99	50.45	33.14	8.15	104.0	7.92
	M	11/24/24 8:17	6.52	17.70	48.65	31.82	8.17	107.0	8.42
	S	11/24/24 8:18	0.73	17.64	48.43	31.66	8.17	106.5	8.40
BI-SDK4	B	11/24/24 8:32	13.16	18.47	49.64	32.54	8.16	105.7	8.16
	M	11/24/24 8:33	6.55	18.43	48.81	31.94	8.16	102.7	7.97
	S	11/24/24 8:34	0.73	18.10	48.50	31.71	8.17	104.1	8.14
BI-DK11	B	11/24/24 8:45	9.72	19.05	50.10	32.88	8.18	109.6	8.35
	M	11/24/24 8:46	4.80	18.14	48.98	32.06	8.17	106.1	8.27
	S	11/24/24 8:47	0.73	17.98	48.88	31.98	8.17	105.5	8.25
BI-NDK3	B	11/24/24 8:58	8.34	18.19	48.95	32.04	8.19	109.9	8.56
	M	11/24/24 8:59	4.22	18.07	48.74	31.89	8.19	110.4	8.63
	S	11/24/24 9:00	0.74	18.03	20.51	12.31	8.19	109.3	8.55
BI-NDK7	B	11/24/24 9:11	10.68	18.78	49.24	32.25	8.17	107.9	8.29
	M	11/24/24 9:12	5.43	17.68	48.12	31.43	8.19	109.7	8.66
	S	11/24/24 9:14	0.74	17.61	47.98	31.33	8.18	109.6	8.66
BI-DK9	B	11/24/24 9:23	9.60	18.78	49.80	32.66	8.21	115.0	8.83
	M	11/24/24 9:24	4.79	18.11	48.80	31.93	8.20	112.1	8.75
	S	11/24/24 9:25	0.74	18.10	48.79	31.92	8.20	111.1	8.67

table 1 continued

BI-SDK6	B	11/24/24 9:35	4.48	18.29	48.70	31.86	8.19	107.7	8.38
	M	11/24/24 9:36	2.11	18.30	48.71	31.86	8.19	108.0	8.40
	S	11/24/24 9:37	0.75	18.31	48.68	31.84	8.19	107.7	8.38
BI-SDK3	B	11/24/24 9:51	7.88	18.17	48.88	31.99	8.20	110.6	8.62
	M	11/24/24 9:53	4.45	18.12	48.77	31.91	8.20	111.0	8.66
	S	11/24/24 9:54	0.75	18.14	48.72	31.87	8.20	110.9	8.65
BI-SDK5	B	11/24/24 10:04	11.69	19.48	50.07	32.86	8.16	106.5	8.05
	M	11/24/24 10:05	6.75	18.97	48.99	32.07	8.17	108.3	8.31
	S	11/24/24 10:07	0.77	18.44	48.16	31.46	8.19	108.8	8.46
BI-DK7	B	11/24/24 10:22	9.07	18.05	48.48	31.69	8.22	112.2	8.78
	M	11/24/24 10:23	4.42	18.05	48.49	31.70	8.21	112.2	8.78
	S	11/24/24 10:24	0.74	18.04	39.33	25.12	8.21	111.8	8.75
BI-NDK2	B	11/24/24 10:36	18.23	19.77	50.79	33.39	8.16	101.5	7.61
	M	11/24/24 10:37	9.15	18.27	48.64	31.81	8.20	110.4	8.59
	S	11/24/24 10:38	0.74	17.84	47.96	31.31	8.19	110.1	8.66
BI-DK5	B	11/24/24 10:50	11.06	19.52	50.06	32.85	8.17	107.0	8.08
	M	11/24/24 10:51	5.65	18.48	48.58	31.77	8.19	108.9	8.44
	S	11/24/24 10:52	0.73	18.47	48.50	31.71	8.19	109.3	8.48
BI-SDK2	B	11/24/24 11:02	13.18	19.82	50.16	32.92	8.16	105.7	7.94
	M	11/24/24 11:04	7.61	19.10	49.24	32.25	8.16	104.4	7.98
	S	11/24/24 11:05	0.74	18.57	48.32	31.58	8.19	108.5	8.41
BI-SDK1	B	11/24/24 11:22	9.65	20.34	50.53	33.19	8.19	109.6	8.14
	M	11/24/24 11:23	4.81	18.89	48.35	31.60	8.18	109.9	8.47
	S	11/24/24 11:24	0.72	18.55	47.80	31.20	8.20	111.9	8.70
BI-DK3	B	11/24/24 11:36	6.11	18.12	47.79	31.19	8.21	112.3	8.80
	M	11/24/24 11:37	3.03	18.12	47.75	31.16	8.21	112.7	8.83
	S	11/24/24 11:38	0.71	18.13	47.76	31.17	8.21	113.2	8.87
BI-NDK6	B	11/24/24 11:52	19.06	19.91	50.90	33.47	8.14	96.1	7.18
	M	11/24/24 11:53	9.26	18.09	48.09	31.41	8.19	108.4	8.49
	S	11/24/24 11:54	0.71	17.88	47.65	31.09	8.19	109.9	8.66
BI-NDK1	B	11/24/24 12:06	16.66	19.60	49.88	32.72	8.14	98.9	7.47
	M	11/24/24 12:07	8.59	19.00	48.96	32.05	8.15	102.4	7.85
	S	11/24/24 12:08	0.69	18.23	47.81	31.20	8.19	108.7	8.50
BI-DK1	B	11/24/24 12:22	5.04	18.18	47.28	30.82	8.22	114.0	8.94
	M	11/24/24 12:23	2.15	18.20	47.29	30.83	8.22	114.6	8.99
	S	11/24/24 12:24	0.69	18.20	47.29	30.83	8.22	114.7	9.00
BI-NDK8	B	11/25/24 8:46	20.79	19.08	49.71	32.60	8.05	99.2	7.57
	M	11/25/24 8:47	10.45	18.59	48.59	31.78	8.11	109.3	8.46
	S	11/25/24 8:48	0.74	17.76	46.45	30.22	8.09	108.5	8.61
BI-NDK5	B	11/25/24 9:02	6.74	18.68	48.59	31.77	8.12	111.6	8.62
	M	11/25/24 9:03	3.53	18.50	48.38	31.63	8.11	109.2	8.47
	S	11/25/24 9:04	0.74	18.57	48.19	31.49	8.10	108.3	8.40

table 1 continued

East Ship Island Sites									
BI-ES26	B	11/23/24 10:57	15.69	18.84	49.94	32.76	8.16	99.8	7.65
	M	11/23/24 10:59	7.66	18.92	49.47	32.42	8.16	101.0	7.74
	S	11/23/24 11:00	0.75	18.81	49.12	32.16	8.16	101.2	7.78
BI-ES25	B	11/23/24 11:15	13.48	18.51	49.81	32.67	8.13	95.1	7.33
	M	11/23/24 11:16	7.56	18.53	48.86	31.97	8.17	101.7	7.87
	S	11/23/24 11:17	0.76	18.57	47.74	31.15	8.16	103.7	8.06
BI-ES24	B	11/23/24 11:28	13.58	19.02	49.68	32.57	8.16	100.1	7.65
	M	11/23/24 11:29	6.64	18.66	48.90	32.00	8.17	102.1	7.88
	S	11/23/24 11:30	0.74	18.81	47.94	31.30	8.16	104.4	8.07
BI-ES20	B	11/23/24 11:40	11.71	19.31	49.78	32.64	8.17	101.4	7.70
	M	11/23/24 11:41	5.96	18.87	49.08	32.14	8.17	101.8	7.82
	S	11/23/24 11:42	0.73	19.03	48.21	31.50	8.16	102.7	7.90
BI-ES21	B	11/23/24 11:53	10.96	19.14	49.88	32.72	8.17	101.2	7.71
	M	11/23/24 11:55	5.15	19.07	48.86	31.98	8.16	102.6	7.86
	S	11/23/24 11:56	0.72	19.09	48.87	31.98	8.16	102.7	7.86
BI-ES19	B	11/24/24 12:35	14.31	19.33	49.45	32.40	8.14	97.5	7.41
	M	11/24/24 12:36	6.49	18.45	47.92	31.29	8.18	105.9	8.25
	S	11/24/24 12:38	0.68	18.30	47.55	31.01	8.19	109.0	8.52
BI-ES18	B	11/24/24 12:49	17.95	19.47	49.83	32.68	8.11	93.7	7.10
	M	11/24/24 12:50	8.78	19.08	49.16	32.20	8.14	98.6	7.54
	S	11/24/24 12:52	0.68	18.01	46.72	30.42	8.20	110.4	8.71
BI-ES23	B	11/24/24 13:07	8.67	20.13	49.25	32.26	8.17	105.1	7.88
	M	11/24/24 13:08	4.00	19.15	48.40	31.64	8.15	102.2	7.84
	S	11/24/24 13:09	0.67	18.49	46.69	30.39	8.19	109.8	8.58
BI-ES22	B	11/24/24 13:18	9.46	20.12	49.29	32.29	8.09	87.9	6.59
	M	11/24/24 13:20	4.65	19.27	47.71	31.13	8.17	104.7	8.03
	S	11/24/24 13:21	0.65	18.82	46.86	30.52	8.20	112.3	8.72
Camille Cut Sites									
BI-CC14	B	11/23/24 9:53	12.49	18.76	49.98	32.79	8.16	99.6	7.64
	M	11/23/24 9:54	6.14	19.16	49.98	32.79	8.16	99.7	7.59
	S	11/23/24 9:55	0.86	19.18	49.98	32.79	8.16	99.7	7.59
BI-CC15	B	11/23/24 10:09	12.66	19.01	49.98	32.79	8.18	100.9	7.70
	M	11/23/24 10:10	6.62	19.19	49.96	32.78	8.16	100.1	7.61
	S	11/23/24 10:11	0.89	19.25	49.94	32.77	8.16	100.2	7.61
BI-CC16	B	11/23/24 10:24	10.39	18.66	50.03	32.83	8.16	99.6	7.65
	M	11/23/24 10:25	6.14	19.18	49.96	32.78	8.16	100.0	7.61
	S	11/23/24 10:26	0.80	19.24	49.96	32.78	8.16	100.0	7.60
BI-CC17	B	11/23/24 10:38	15.47	18.90	49.96	32.78	8.17	100.8	7.71
	M	11/23/24 10:39	7.55	19.28	49.97	32.79	8.16	100.6	7.64
	S	11/23/24 10:40	0.76	19.39	49.97	32.78	8.16	100.7	7.63

table 1 continued

BI-CC13	B	11/24/24 13:35	14.97	19.82	49.53	32.46	8.13	95.7	7.21
	M	11/24/24 13:36	7.26	19.60	49.23	32.25	8.18	108.3	8.20
	S	11/24/24 13:37	0.64	19.48	47.13	30.71	8.21	112.5	8.61
BI-CC8	B	11/24/24 13:47	18.21	19.68	49.67	32.56	8.12	94.3	7.12
	M	11/24/24 13:48	8.94	19.68	49.65	32.55	8.13	96.9	7.31
	S	11/24/24 13:50	0.65	19.41	47.32	30.85	8.19	109.6	8.40
BI-CC12	B	11/24/24 14:05	15.82	19.70	49.64	32.55	8.15	100.4	7.58
	M	11/24/24 14:07	7.76	19.74	49.60	32.51	8.15	102.8	7.75
	S	11/24/24 14:08	0.65	19.57	47.79	31.19	8.19	111.8	8.52
BI-CC7	B	11/25/24 9:29	17.68	19.71	49.85	32.69	8.06	93.7	7.07
	M	11/25/24 9:31	8.80	19.44	48.96	32.05	8.08	98.2	7.47
	S	11/25/24 9:32	0.75	18.19	0.83	0.41	8.14	108.0	10.16
BI-CC11	B	11/25/24 9:45	17.20	19.60	49.81	32.67	8.07	96.4	7.28
	M	11/25/24 9:46	8.61	19.17	48.58	31.77	8.10	100.0	7.66
	S	11/25/24 9:47	0.74	18.61	47.17	30.74	8.13	106.9	8.32
BI-CC6	B	11/25/24 9:59	16.62	19.63	49.84	32.69	8.09	98.4	7.43
	M	11/25/24 10:00	8.38	19.59	49.64	32.55	8.10	100.3	7.58
	S	11/25/24 10:01	0.74	18.71	47.07	30.67	8.14	106.6	8.28
BI-CC10	B	11/25/24 10:12	17.79	19.54	49.80	32.66	8.09	96.1	7.27
	M	11/25/24 10:13	8.61	19.37	49.12	32.16	8.11	100.4	7.64
	S	11/25/24 10:14	0.73	18.97	47.58	31.04	8.13	104.3	8.05
BI-CC9	B	11/25/24 10:26	17.90	19.41	49.77	32.64	8.08	94.0	7.13
	M	11/25/24 10:27	8.99	19.18	48.93	32.03	8.15	105.3	8.05
	S	11/25/24 10:29	0.74	19.24	48.09	31.41	8.14	104.4	8.00
BI-CC5	B	11/25/24 10:43	18.39	19.42	49.83	32.68	8.11	99.8	7.56
	M	11/25/24 10:44	9.12	19.27	48.39	31.63	8.12	100.4	7.68
	S	11/25/24 10:45	0.73	19.19	47.53	31.00	8.13	104.7	8.05
West Ship Island Sites									
BI-WS9	B	10/28/24 8:56	19.67	23.35	49.21	32.19	7.92	85.9	6.08
	M	10/28/24 8:58	9.74	23.64	48.44	31.62	7.99	98.0	6.93
	S	10/28/24 8:59	0.79	23.58	47.77	31.14	8.01	97.4	6.91
BI-WS8	B	10/28/24 9:17	21.22	23.34	49.28	32.24	8.08	89.5	6.34
	M	10/28/24 9:18	11.35	23.41	49.20	32.18	8.11	94.0	6.65
	S	10/28/24 9:19	0.81	23.69	48.03	31.32	8.10	97.2	6.88
BI-WS7	B	10/28/24 9:35	21.42	23.36	49.31	32.26	8.10	91.2	6.46
	M	10/28/24 9:36	10.67	23.55	48.76	31.86	8.13	96.2	6.80
	S	10/28/24 9:37	0.80	23.56	47.99	31.30	8.14	100.8	7.15
BI-WS3	B	10/28/24 9:55	21.38	23.34	49.32	32.27	8.13	91.2	6.46
	M	10/28/24 9:56	10.50	23.54	48.52	31.68	8.15	98.3	6.96
	S	10/28/24 9:57	0.81	23.57	47.95	31.27	8.15	100.6	7.13

table 1 continued

BI-WS4	B	10/28/24 11:53	16.26	23.43	49.24	32.21	8.15	93.7	6.63
	M	10/28/24 11:54	8.25	23.59	48.94	31.99	8.17	98.6	6.96
	S	10/28/24 11:55	0.80	24.02	48.25	31.47	8.18	103.8	7.29
BI-WS1	B	10/28/24 12:11	27.30	23.40	49.68	32.53	8.12	81.9	5.78
	M	10/28/24 12:12	13.60	23.45	49.23	32.21	8.15	91.5	6.47
	S	10/28/24 12:14	0.80	23.82	42.92	27.63	8.17	103.9	7.49
BI-WS2	B	10/28/24 12:37	24.97	23.44	49.70	32.55	8.14	88.5	6.25
	M	10/28/24 12:38	12.41	23.66	48.69	31.81	8.18	100.6	7.10
	S	10/28/24 12:39	0.80	23.91	42.95	27.65	8.17	104.7	7.54
BI-WS6	B	11/23/24 8:20	18.73	19.71	49.87	32.71	8.07	94.6	7.13
	M	11/23/24 8:22	8.90	18.87	47.14	30.72	8.13	98.2	7.61
	S	11/23/24 8:23	0.79	18.32	45.73	29.70	8.16	101.6	8.01
BI-WS5	B	11/23/24 8:43	13.55	19.44	49.80	32.66	8.14	97.0	7.35
	M	11/23/24 8:44	6.95	18.75	47.26	30.80	8.16	99.2	7.70
	S	11/23/24 8:45	0.76	16.90	0.56	0.27	8.16	99.9	9.65
Ship Island Pass Sites									
GS-2	B	10/25/24 8:21	12.03	22.47	48.39	31.60	7.84	87.8	6.34
	M	10/25/24 8:22	6.00	22.47	48.36	31.58	7.92	88.6	6.40
	S	10/25/24 8:24	0.75	21.96	40.40	25.86	7.97	100.3	7.55
GS-3	B	10/25/24 8:53	13.88	22.52	48.51	31.69	8.05	89.6	6.45
	M	10/25/24 8:54	6.89	22.52	48.48	31.67	8.07	90.1	6.49
	S	10/25/24 8:55	0.76	22.16	39.99	25.56	8.09	101.6	7.63
GS-5	B	10/25/24 9:09	14.73	22.639	48.581	31.74	8.1	91.6	6.59
	M	10/25/24 9:10	7.355	22.759	47.763	31.14	8.13	102	7.34
	S	10/25/24 9:12	0.77	22.033	41.414	26.58	8.13	102.4	7.67
BI-SIP3	B	10/25/24 9:31	18.44	22.70	48.78	31.89	8.13	91.9	6.59
	M	10/25/24 9:32	9.30	22.70	48.11	31.39	8.15	100.1	7.21
	S	10/25/24 9:34	0.77	22.55	47.22	30.75	8.16	103.6	7.50
BI-SIP4	B	10/25/24 9:47	16.92	22.52	48.66	31.80	8.13	88.0	6.34
	M	10/25/24 9:50	8.64	22.60	48.63	31.77	8.14	92.2	6.63
	S	10/25/24 9:51	0.78	22.20	42.53	27.37	8.16	102.2	7.59
BI-SIP6	B	10/25/24 10:09	11.54	22.46	48.17	31.44	8.13	90.7	6.55
	M	10/25/24 10:10	5.79	22.30	47.52	30.97	8.14	94.8	6.89
	S	10/25/24 10:12	0.77	22.27	44.97	29.13	8.15	98.7	7.25
GS-22	B	10/25/24 10:28	14.38	22.47	48.45	31.65	8.13	86.5	6.24
	M	10/25/24 10:30	7.36	22.46	48.26	31.51	8.14	90.2	6.52
	S	10/25/24 10:31	0.77	22.42	46.94	30.55	8.15	98.2	7.14
GS-23	B	10/25/24 10:42	18.48	22.56	48.64	31.79	8.14	89.2	6.42
	M	10/25/24 10:43	8.40	22.60	48.51	31.69	8.16	94.4	6.79
	S	10/25/24 10:44	0.78	22.53	46.35	30.12	8.16	101.1	7.35

table 1 continued

GS-25	B	10/25/24 11:05	21.98	22.65	48.92	31.99	8.14	90.5	6.50
	M	10/25/24 11:06	10.70	22.69	48.30	31.54	8.17	98.7	7.10
	S	10/25/24 11:07	0.78	22.42	42.67	27.47	8.18	105.4	7.80
GS-28	B	10/28/24 10:15	22.34	23.40	49.88	32.68	8.10	80.6	5.68
	M	10/28/24 10:16	11.06	23.53	48.83	31.91	8.17	100.5	7.11
	S	10/28/24 10:17	0.81	23.63	47.35	30.82	8.16	102.7	7.30
GS-27	B	10/28/24 10:33	22.02	23.33	49.77	32.60	8.10	78.1	5.52
	M	10/28/24 10:34	11.31	23.39	49.45	32.36	8.15	90.5	6.40
	S	10/28/24 10:35	0.82	23.52	43.90	28.33	8.17	103.0	7.43
BI-SIP2	B	10/28/24 10:54	20.36	23.24	49.33	32.28	8.13	86.4	6.13
	M	10/28/24 10:55	10.11	23.37	49.12	32.12	8.16	93.7	6.63
	S	10/28/24 10:56	0.83	23.39	42.36	27.23	8.17	104.2	7.58
BI-SIP1	B	10/28/24 11:12	19.89	23.47	49.36	32.29	8.15	91.2	6.44
	M	10/28/24 11:14	10.15	23.56	48.87	31.94	8.17	97.4	6.88
	S	10/28/24 11:15	0.84	23.68	44.34	28.65	8.17	103.8	7.46
GS-20	B	10/28/24 11:33	17.06	23.54	49.19	32.17	8.17	95.6	6.75
	M	10/28/24 11:34	8.06	23.70	48.22	31.46	8.18	103.2	7.29
	S	10/28/24 11:36	0.81	23.68	43.10	27.76	8.17	104.5	7.54
GS-10	B	11/23/24 9:01	20.22	19.45	49.79	32.65	8.14	95.5	7.24
	M	11/23/24 9:03	10.19	18.66	47.71	31.13	8.15	97.7	7.58
	S	11/23/24 9:04	0.76	18.39	46.52	30.27	8.16	100.3	7.86
GS-9	B	11/23/24 9:15	16.90	19.53	49.82	32.67	8.15	97.8	7.39
	M	11/23/24 9:16	8.32	18.43	46.99	30.61	8.16	100.1	7.83
	S	11/23/24 9:17	0.76	18.29	46.70	30.40	8.16	100.8	7.91
Borrow Sites									
BS1	B	10/25/24 11:37	15.84	22.53	48.49	31.68	8.14	89.8	6.47
	M	10/25/24 11:38	8.11	22.46	47.22	30.75	8.16	97.3	7.06
	S	10/25/24 11:39	0.78	23.01	40.26	25.74	8.16	106.0	7.84
BS2	B	10/25/24 11:54	18.67	22.53	48.47	31.66	8.13	88.6	6.38
	M	10/25/24 11:56	9.84	22.51	48.40	31.61	8.14	90.5	6.52
	S	10/25/24 11:57	0.77	23.25	39.19	24.98	8.16	106.9	7.90
BS3	B	10/25/24 12:11	18.75	22.54	48.58	31.74	8.13	87.1	6.27
	M	10/25/24 12:12	9.75	22.53	48.45	31.64	8.14	90.5	6.52
	S	10/25/24 12:14	0.76	23.42	39.32	25.07	8.15	107.7	7.94
BS4	B	10/25/24 12:27	17.72	22.57	48.55	31.72	8.13	90.9	6.54
	M	10/25/24 12:28	9.03	22.59	48.46	31.65	8.15	92.7	6.67
	S	10/25/24 12:30	0.76	23.49	39.27	25.03	8.16	108.7	8.00
BS5	B	10/25/24 12:45	17.52	22.57	48.55	31.72	8.12	89.0	6.41
	M	10/25/24 12:46	8.90	22.61	47.42	30.90	8.15	95.4	6.90
	S	10/25/24 12:47	0.77	23.27	38.65	24.60	8.16	109.4	8.11

table 1 continued

BS6	B	10/25/24 13:02	16.95	22.59	48.55	31.72	8.13	90.8	6.54
	M	10/25/24 13:03	10.13	22.59	47.90	31.24	8.16	98.7	7.13
	S	10/25/24 13:05	0.75	23.57	39.49	25.19	8.16	110.4	8.10

Figure 1. Overall map (and map key designations) for the gulf sturgeon foraging habitat sampling station locations.

Figure 2.

GS_2
GS_3
GS_5
BS6
BS5
BI_SIP6
BS4
BS3
BI_SIP4
BS2
BI_SIP3
BS1
GS_22
GS_23
GS_25
MAP 1
0
2,500
5,000
10,000
Feet
N
Service Layer Credits: Source: Esri, Maxar, Earthstar Geographics, and the GIS User Community

Figure 3.

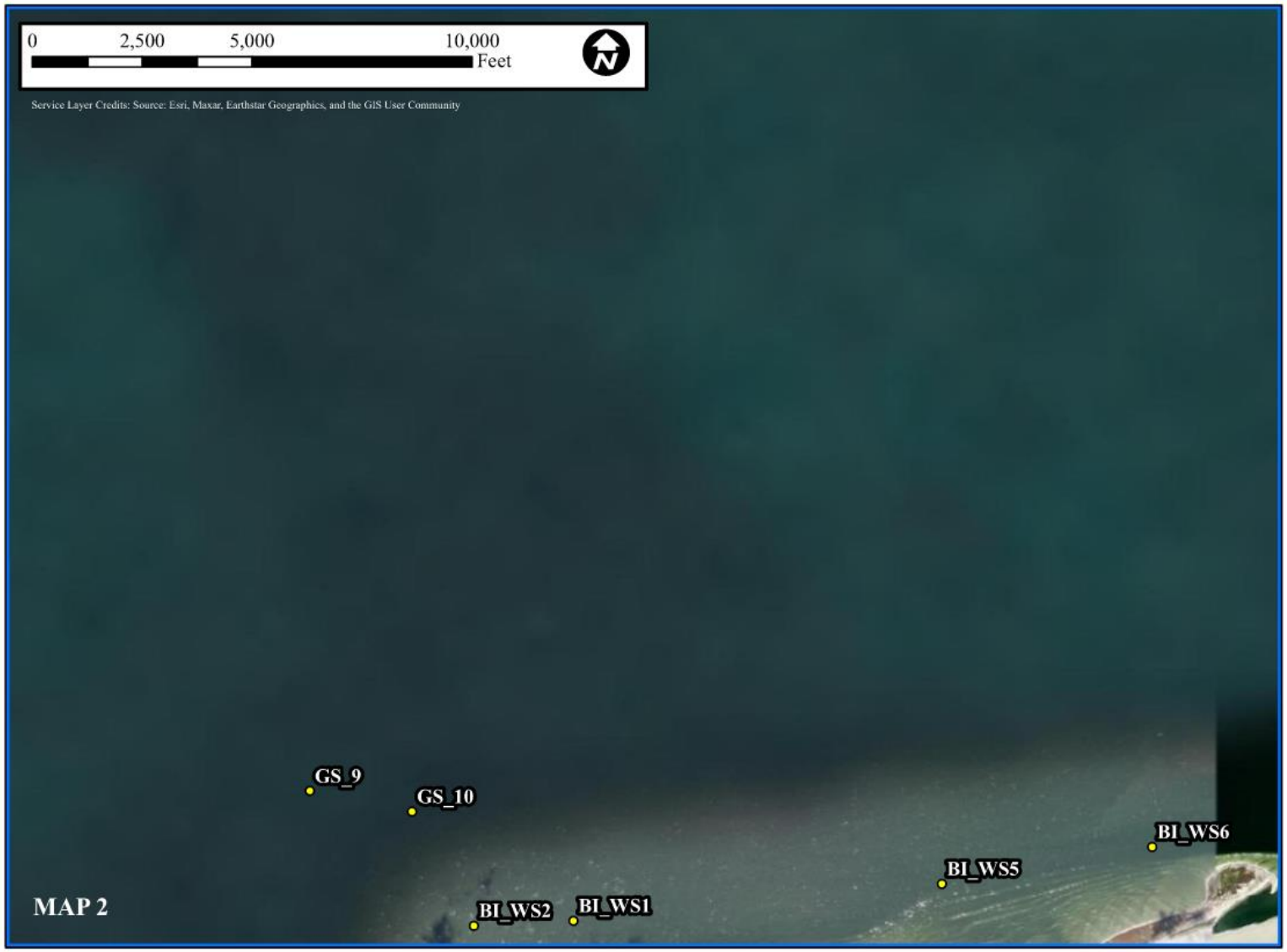
0
2,500
5,000
10,000
Feet
N
Service Layer Credits: Source: Esri, Maxar, Earthstar Geographics, and the GIS User Community
GS_9
GS_10
BI_WS6
BI_WS5
BI_WS2
BI_WS1
MAP 2

Figure 4.

BI_SIP2
BI_SIP1
GS_20
BI_WS4
BI_WS3
BI_WS7
BI_WS8
BI_WS9
GS_27
GS_28
MAP 3
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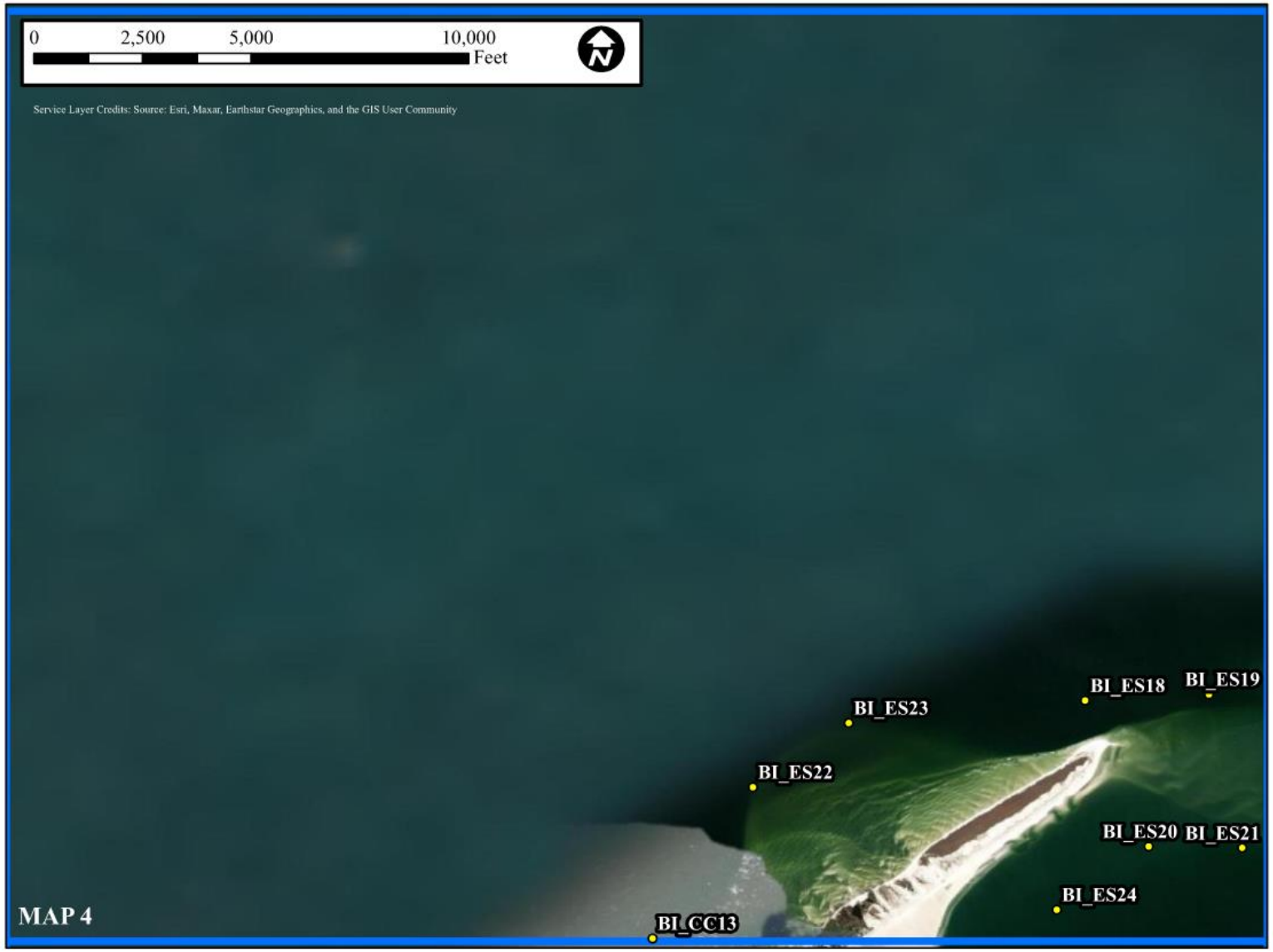
2,500
5,000
10,000
Feet
N
Service Layer Credits: Source: Esri, Maxar, Earthstar Geographics, and the GIS User Community

Figure 5.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2,500
5,000
10,000
Feet
N
Service Layer Credits: Source: Esri, Maxar, Earthstar Geographics, and the GIS User Community
BI_ES18
BI_ES19
BI_ES23
BI_ES22
BI_ES20
BI_ES21
BI_ES24
BI_CC13
MAP 4

Figure 6.

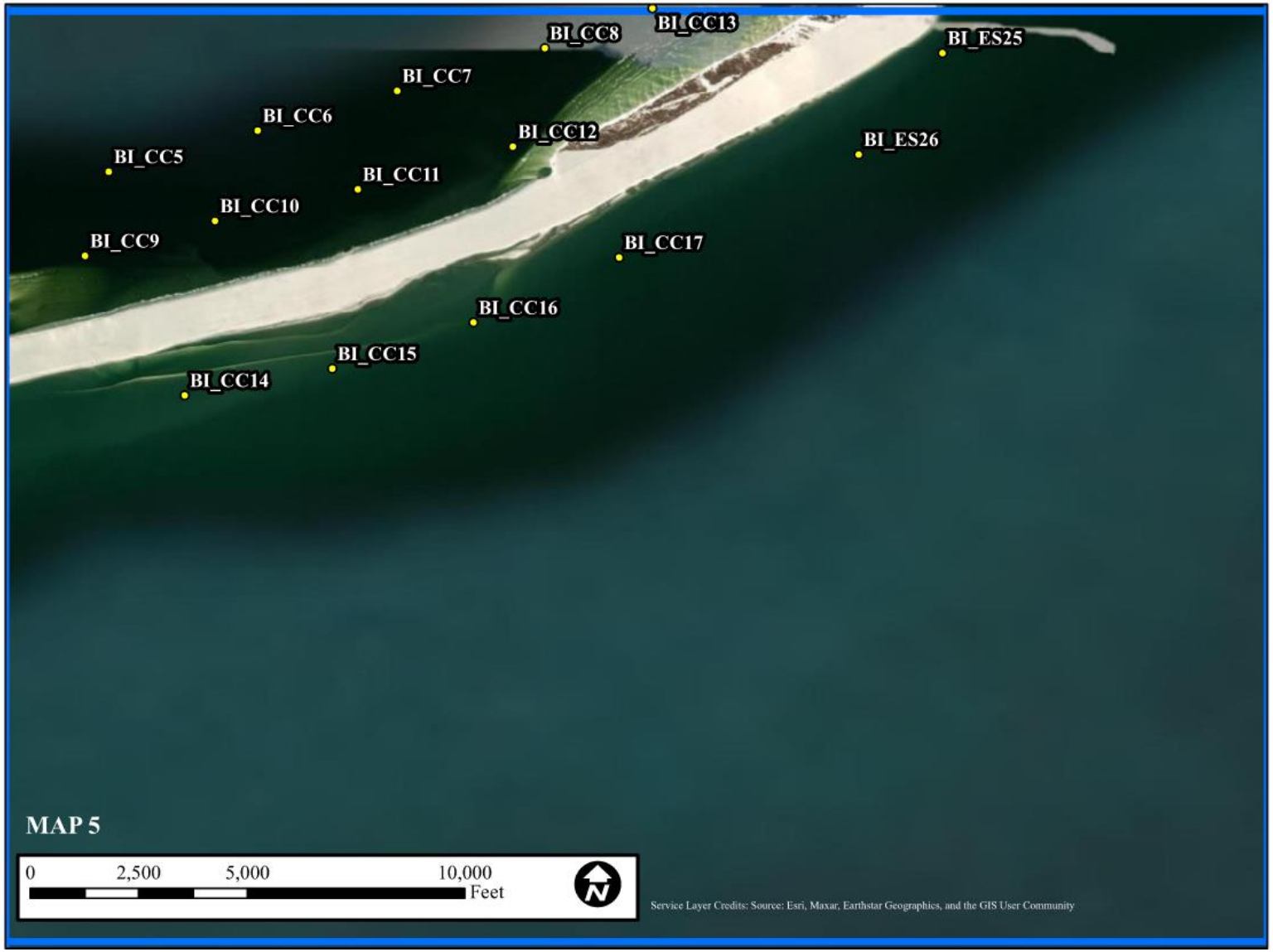

BI_CC13
BI_CC8
BI_ES25
BI_CC7
BI_CC6
BI_CC12
BI_ES26
BI_CC5
BI_CC11
BI_CC10
BI_CC9
BI_CC17
BI_CC16
BI_CC15
BI_CC14
MAP 5
0
2,500
5,000
10,000
Feet
N
Service Layer Credits: Source: Esri, Maxar, Earthstar Geographics, and the GIS User Community

Figure 7.

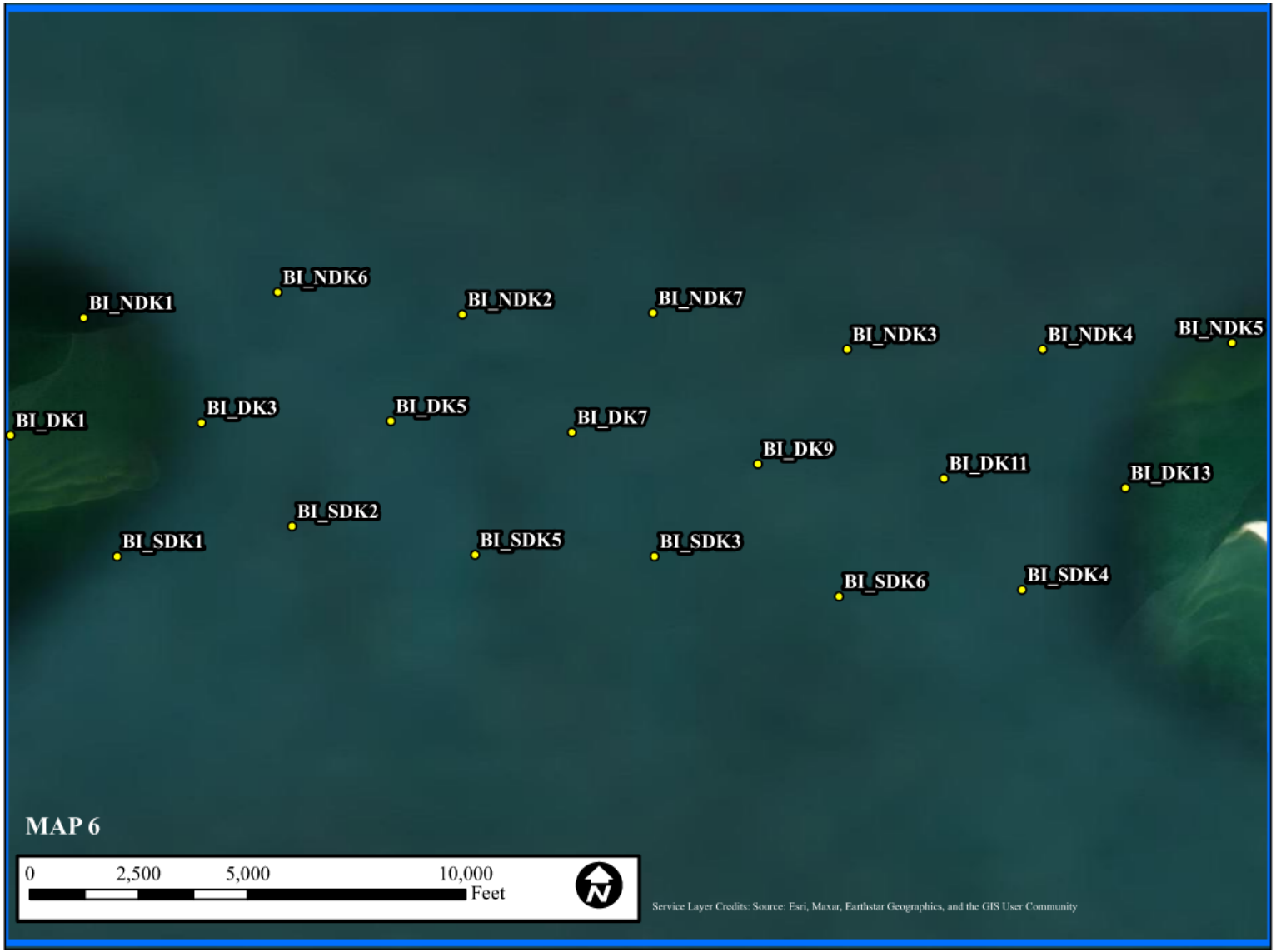
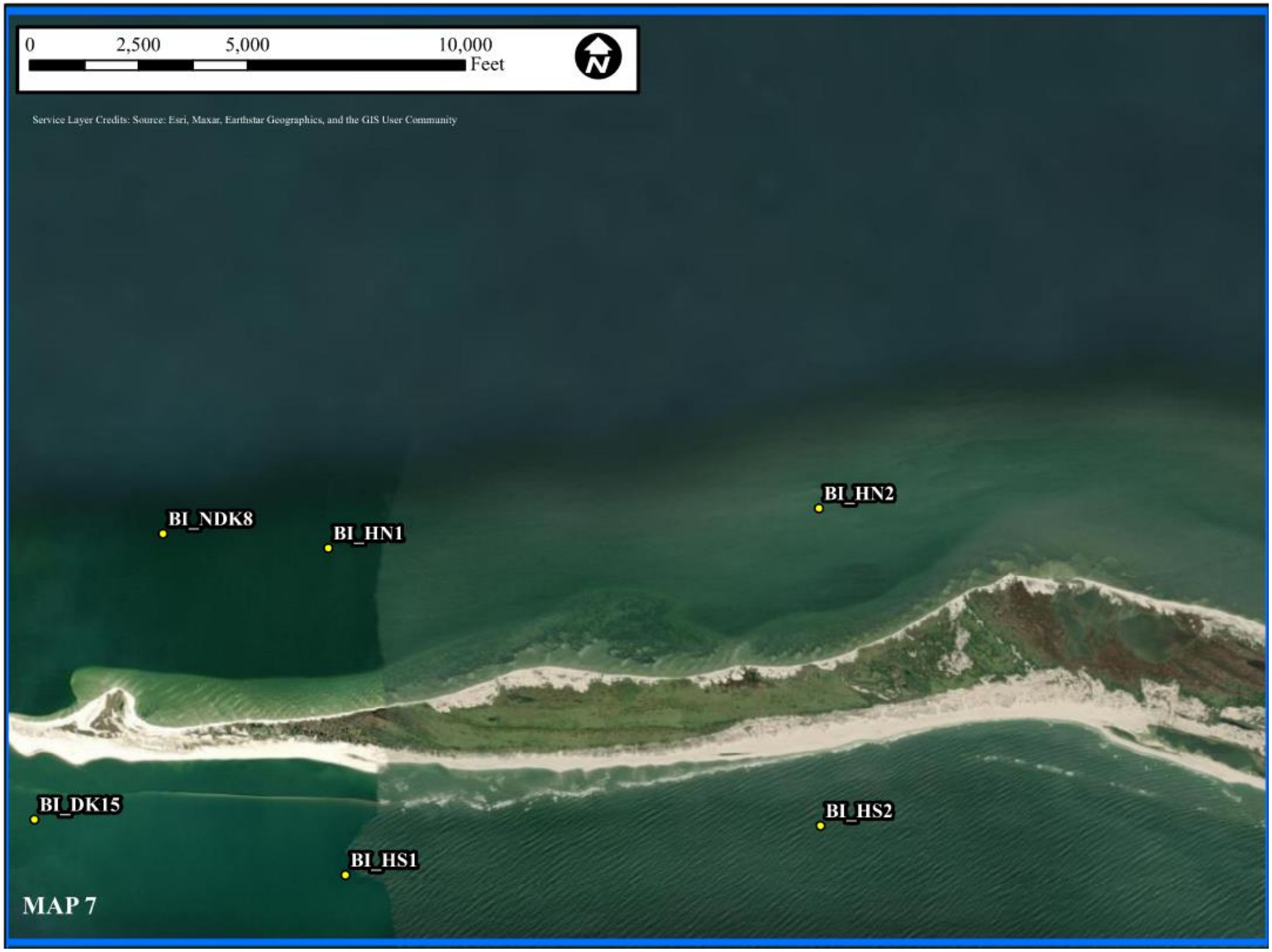

BI_NDK1
BI_NDK6
BI_NDK2
BI_NDK7
BI_NDK3
BI_NDK4
BI_NDK5
BI_DK1
BI_DK3
BI_DK5
BI_DK7
BI_DK9
BI_DK11
BI_DK13
BI_SDK2
BI_SDK1
BI_SDK5
BI_SDK3
BI_SDK6
BI_SDK4
MAP 6
0
2,500
5,000
10,000
Feet
N
Service Layer Credits: Source: Esri, Maxar, Earthstar Geographics, and the GIS User Community

Figure 8.


0
2,500
5,000
10,000
Feet
N
Service Layer Credits: Source: Esri, Maxar, Earthstar Geographics, and the GIS User Community
BI_NDK8
BI_HN1
BI_HN2
BI_DK15
BI_HS1
BI_HS2
MAP 7